Formula Sheet

CBSE Class 12 Mathematics • NCERT Part I, Chapters 1–6

Blue boxes are NCERT (board) formulas. Purple **JEE extra** boxes go beyond the NCERT book and are not needed for the board exam. Use this sheet with the separate revision notes.

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Chapter 1 Relations and Functions

Relations

Empty relation	$R = \phi$ Universal: $R = A \times A$
Reflexive	$(a, a) \in R$ for every $a \in A$
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$
Transitive	$(a, b), (b, c) \in R \Rightarrow (a, c) \in R$
Equivalence	reflexive + symmetric + transitive; classes $[a] = \{x : (x, a) \in R\}$ are disjoint and cover A

Functions

One-one (injective): $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. **Onto (surjective):** range = co-domain.
Bijective: one-one and onto. Finite set X: $f : X \to X$ one-one $\iff$ onto.
$[x]$: greatest integer $\leq x$; $|x|$; signum: $1, 0, -1$ for $x > 0, = 0, < 0$.

JEE extra

$|A| = m$, $|B| = n$: relations 2^{mn}; functions n^m; one-one $\frac{n!}{(n-m)!}$; bijections $(m = n)$ $n!$. Reflexive relations on n elements: 2^{n^2-n}; symmetric: $2^{n(n+1)/2}$.
$(g \circ f)(x) = g(f(x))$; f invertible $\iff$ bijective; $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Chapter 2 Inverse Trigonometric Functions

Domains and principal value ranges

$\sin^{-1} x$	$[-1,1]$	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$	$\operatorname{cosec}^{-1} x$	$\mathbf{R}-(-1,1)$	$\left[-\frac{\pi}{2},\frac{\pi}{2}\right]-\{0\}$
$\cos^{-1} x$	$[-1,1]$	$[0,\pi]$	$\sec^{-1} x$	$\mathbf{R}-(-1,1)$	$[0,\pi]-\left\{\frac{\pi}{2}\right\}$
$\tan^{-1} x$	$\mathbf{R}$	$\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$	$\cot^{-1} x$	$\mathbf{R}$	$(0,\pi)$

Properties

$\sin(\sin^{-1} x)=x,\ x\in[-1,1]$; $\sin^{-1}(\sin x)=x,\ x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (similar for others in their ranges).
$\sin^{-1}(-x)=-\sin^{-1} x$, $\tan^{-1}(-x)=-\tan^{-1} x$, $\operatorname{cosec}^{-1}(-x)=-\operatorname{cosec}^{-1} x$
$\cos^{-1}(-x)=\pi-\cos^{-1} x$, $\sec^{-1}(-x)=\pi-\sec^{-1} x$, $\cot^{-1}(-x)=\pi-\cot^{-1} x$
$\sin^{-1}\frac{1}{x}=\operatorname{cosec}^{-1} x$, $\cos^{-1}\frac{1}{x}=\sec^{-1} x$, $\tan^{-1}\frac{1}{x}=\cot^{-1} x\ (x>0)$
Useful values: $\sin^{-1}\frac{1}{2}=\frac{\pi}{6}$, $\cos^{-1}\frac{1}{2}=\frac{\pi}{3}$, $\tan^{-1}1=\frac{\pi}{4}$, $\tan^{-1}\sqrt{3}=\frac{\pi}{3}$, $\sin^{-1}\frac{1}{\sqrt{2}}=\frac{\pi}{4}$.

JEE extra

$\sin^{-1} x+\cos^{-1} x=\tan^{-1} x+\cot^{-1} x=\sec^{-1} x+\operatorname{cosec}^{-1} x=\frac{\pi}{2}$.

$\tan^{-1} x\pm\tan^{-1} y=\tan^{-1}\dfrac{x\pm y}{1\mp xy}$. $2\tan^{-1} x=\sin^{-1}\dfrac{2x}{1+x^2}=\cos^{-1}\dfrac{1-x^2}{1+x^2}=\tan^{-1}\dfrac{2x}{1-x^2}$ $(|x|<1)$.

$3\sin^{-1} x=\sin^{-1}(3x-4x^3)$, $3\cos^{-1} x=\cos^{-1}(4x^3-3x)$, $3\tan^{-1} x=\tan^{-1}\dfrac{3x-x^3}{1-3x^2}$.

Chapter 3 Matrices

Types and operations

Order $m\times n$; square $(m=n)$; diagonal $(a_{ij}=0,\ i\neq j)$; scalar $(a_{ii}=k)$; identity I; zero O; row $1\times n$; column $m\times 1$.

$A+B$, kA: entry-wise (same order). AB: defined if columns of A = rows of B; $(AB)_{ik}=\sum_j a_{ij}b_{jk}$; order $m\times p$.

$AB\neq BA$ in general; $AB=O$ does not imply $A=O$ or $B=O$. $A(BC)=(AB)C$, $A(B+C)=AB+AC$.

Transpose, symmetric, skew-symmetric, inverse

$(A')'=A$, $(kA)'=kA'$, $(A+B)'=A'+B'$, $(AB)'=B'A'$.

Symmetric: $A'=A$; skew-symmetric: $A'=-A$ (diagonal entries 0).

Every square matrix: $A=\underbrace{\tfrac{1}{2}(A+A')}_{\text{symmetric}}+\underbrace{\tfrac{1}{2}(A-A')}_{\text{skew}}$.

Inverse: $AB=BA=I\Rightarrow B=A^{-1}$; unique; $(AB)^{-1}=B^{-1}A^{-1}$.

JEE extra

Number of $m \times n$ matrices with k possible entries: k^{mn}. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Idempotent $A^2 = A$; involutory $A^2 = I$; nilpotent $A^k = O$; orthogonal $AA' = I$.

$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}.$$

Chapter 4 Determinants

Evaluation and area

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$; 3×3: $a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$.

$|kA| = k^n|A|$ (n = order); $|AB| = |A||B|$; $|A'| = |A|$.

Area of triangle: $\Delta = \frac{1}{2}\left|\begin{matrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{matrix}\right|$; collinear $\iff \Delta = 0$; line through two points: put (x, y) as third vertex, $\Delta = 0$.

Minors, cofactors, adjoint, inverse

M_{ij}: delete row i, column j; $A_{ij} = (-1)^{i+j}M_{ij}$. $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$ (any row/column). Row $\times$ cofactors of another row $= 0$.

$\operatorname{adj}A = [A_{ij}]'$; 2×2: $\operatorname{adj}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

$A(\operatorname{adj}A) = (\operatorname{adj}A)A = |A|I$; $A^{-1} = \frac{1}{|A|}\operatorname{adj}A$ (exists $\iff |A| \neq 0$, non-singular).

$|\operatorname{adj}A| = |A|^{n-1}$; $|A^{-1}| = \frac{1}{|A|}$.

System of linear equations $AX = B$

$\lvert A\rvert \neq 0$	consistent, unique solution $X = A^{-1}B$
$\lvert A\rvert = 0$, $(\operatorname{adj}A)B \neq O$	inconsistent (no solution)
$\lvert A\rvert = 0$, $(\operatorname{adj}A)B = O$	infinitely many solutions or none

JEE extra

Properties: swapping rows changes the sign; equal/proportional rows $\Rightarrow 0$; $R_i \to R_i + kR_j$ keeps the value. $\operatorname{adj}(\operatorname{adj}A) = |A|^{n-2}A$, $|\operatorname{adj}(\operatorname{adj}A)| = |A|^{(n-1)^2}$.

Cramer: $x = \frac{\Delta_1}{\Delta}$, $y = \frac{\Delta_2}{\Delta}$, $z = \frac{\Delta_3}{\Delta}$.

Chapter 5 Continuity and Differentiability

Continuity and differentiability

Continuous at a: $\lim_{x\to a^-} f = \lim_{x\to a^+} f = f(a)$. Sum, difference, product, quotient ($g \neq 0$), composite of continuous functions are continuous.

Differentiable at a: LHD $= \lim_{h\to 0^-} \dfrac{f(a+h)-f(a)}{h} =$ RHD. Differentiable $\Rightarrow$ continuous (not conversely: $|x|$ at 0).

Derivatives

$\dfrac{d}{dx}x^n = nx^{n-1}$

$\dfrac{d}{dx}\sin x = \cos x,\ \dfrac{d}{dx}\cos x = -\sin x$

$\dfrac{d}{dx}\tan x = \sec^2 x,\ \dfrac{d}{dx}\cot x = -\text{cosec}^2 x$

$\dfrac{d}{dx}\sec x = \sec x\tan x$

$\dfrac{d}{dx}\text{cosec}\, x = -\text{cosec}\, x\cot x$

$\dfrac{d}{dx}e^x = e^x,\ \dfrac{d}{dx}a^x = a^x\log a$

$\dfrac{d}{dx}\log x = \frac{1}{x}$

$\dfrac{d}{dx}\sin^{-1}x = \dfrac{1}{\sqrt{1-x^2}},\ \dfrac{d}{dx}\cos^{-1}x = \dfrac{-1}{\sqrt{1-x^2}}$

$\dfrac{d}{dx}\tan^{-1}x = \frac{1}{1+x^2},\ \dfrac{d}{dx}\cot^{-1}x = \dfrac{-1}{1+x^2}$

$\dfrac{d}{dx}\sec^{-1}x = \dfrac{1}{x\sqrt{x^2-1}},\ \dfrac{d}{dx}\text{cosec}^{-1}x = \dfrac{-1}{x\sqrt{x^2-1}}$

Rules and methods

Product: $(uv)' = u'v + uv'$; quotient: $\left(\frac{u}{v}\right)' = \frac{u'v-uv'}{v^2}$; chain: $\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx}$.

Logarithmic: $y = u^v \Rightarrow \dfrac{dy}{dx} = u^v\left[\dfrac{v}{u}u' + v'\log u\right]$; $\frac{d}{dx}x^x = x^x(1+\log x)$.

Parametric: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$; $\dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)\Big/\dfrac{dx}{dt}$.

Implicit: differentiate both sides, collect $\frac{dy}{dx}$. **Second order:** $y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$.

Simplifying substitutions: $\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2} \to x = \tan\theta$; $2x\sqrt{1-x^2} \to x = \sin\theta$; $2x^2-1 \to x = \cos\theta$.

JEE extra

$\lim_{x\to 0}\dfrac{\sin x}{x} = 1,\ \lim_{x\to 0}\dfrac{e^x-1}{x} = 1,\ \lim_{x\to 0}\dfrac{\log(1+x)}{x} = 1,\ \lim_{x\to 0}\dfrac{1-\cos x}{x^2} = \dfrac{1}{2}$.

Rolle: $f(a) = f(b) \Rightarrow f'(c) = 0$. MVT: $f'(c) = \frac{f(b)-f(a)}{b-a}$. $\frac{d}{dx}|x| = \frac{x}{|x|}$ $(x \neq 0)$. $\frac{d}{dx}\log_a x = \frac{1}{x\log a}$.

Chapter 6 Application of Derivatives

Rate of change

$\frac{dy}{dx}$ (or $f'(x)$) = rate of change of y with respect to x; at a point: $\left.\frac{dy}{dx}\right|_{x=x_0} = f'(x_0)$.

Chain rule (related rates): $\frac{dy}{dt} = \frac{dy}{dx}\cdot\frac{dx}{dt}$, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ $(\frac{dx}{dt} \neq 0)$.

Marginal cost $\text{MC} = \frac{dC}{dx}$; **Marginal revenue** $\text{MR} = \frac{dR}{dx}$; Profit $P = R - C$.

Increasing quantity: $\frac{d}{dt} > 0$; decreasing quantity: $\frac{d}{dt} < 0$ (put a minus sign!).

Mensuration formulas you will need

Shape	Area / Surface	Volume
Circle (radius r)	$A = \pi r^2$, circumference $2\pi r$	–
Square / cube (side x)	$A = x^2$; cube surface $6x^2$	$V = x^3$
Cuboid (l, b, h)	$2(lb + bh + hl)$; open box: $lb + 2h(l + b)$	$V = lbh$
Sphere (radius r)	$S = 4\pi r^2$	$V = \frac{4}{3}\pi r^3$
Cylinder (r, h)	curved $2\pi rh$; total $2\pi r(r + h)$; open $\pi r^2 + 2\pi rh$	$V = \pi r^2 h$
Cone $(r, h$, slant $l)$	curved πrl, $l = \sqrt{r^2 + h^2}$; total $\pi r(r + l)$	$V = \frac{1}{3}\pi r^2 h$

Increasing and decreasing functions

On an interval I (with f continuous on $[a, b]$, differentiable on (a, b)):

$f'(x) > 0$ for all x in (a, b) f is **strictly increasing** on $[a, b]$

$f'(x) < 0$ for all x in (a, b) f is **strictly decreasing** on $[a, b]$

$f'(x) = 0$ for all x in (a, b) f is **constant**

$f'(x) \geq 0$ with $f' = 0$ only at isolated points still gives *increasing* (e.g. x^3).
Method: solve $f'(x) = 0 \to$ split the number line at these points $\to$ check the sign of f' in each part.

Maxima and minima

Critical point: c where $f'(c) = 0$ or f is not differentiable at c.

First derivative test (as x increases through c):
f' changes $+ \to -$: **local maximum**; $- \to +$: **local minimum**; no change of sign: **point of inflexion**.

Second derivative test (with $f'(c) = 0$):
$f''(c) < 0$: local maximum; $f''(c) > 0$: local minimum; $f''(c) = 0$: test fails, use the first derivative test.

Absolute max/min on $[a, b]$**:** find all critical points in (a, b), evaluate f at them *and* at a, b; the largest value is the absolute maximum, the smallest is the absolute minimum.

A continuous function on a closed interval always has an absolute maximum and minimum. Local max value can be less than a local min value.

Useful standard results

- $x+y=k$ (constant): product xy is maximum when $x=y=\frac{k}{2}$; sum $x+y$ with $xy=k$ is minimum when $x=y=\sqrt{k}$.
- $a\sin x+b\cos x$ has maximum $\sqrt{a^2+b^2}$ and minimum $-\sqrt{a^2+b^2}$; $-1\le\sin x,\cos x\le 1$.
- $(x-a)^2+k\ge k$ (min k at $x=a$); $-(x-a)^2+k\le k$ (max k at $x=a$); $|x-a|+k\ge k$.
- Distance: to optimise $\sqrt{D}$, optimise D (same point). To optimise $\log f$ or f^2, same point as f (when $f>0$).

JEE extra

Tangent: $y-y_1=m(x-x_1)$, $m=\frac{dy}{dx}$; normal: slope $-\frac{1}{m}$. Approximation: $f(x+\Delta x)\approx f(x)+f'(x)\Delta x$.
Max of $x^m y^n$ with $x+y=k$: $\frac{x}{m}=\frac{y}{n}$. Cylinder in sphere: $h=\frac{2R}{\sqrt{3}}$; cone in sphere: $h=\frac{4R}{3}$, $V=\frac{8}{27}V_{sphere}$; closed can, least surface: $h=2r$.